COMPRESSED AIR SYSTEM

Prepared by : Hassan alqarni
Abdullah alsubahi

Compressed air system supplies instrument and plant air to the entire phosphate chemical complex and will consist of:

four air compressors of screw type with separate air intake inter cooler and after cooler.

-a separate piping headers for each instrument and plant air .

-two pre-filter. ▣

- two instrument air dryers ▣

(100% duty each) and one electric ▣ heater for regeneration .

two after-filter . ▣

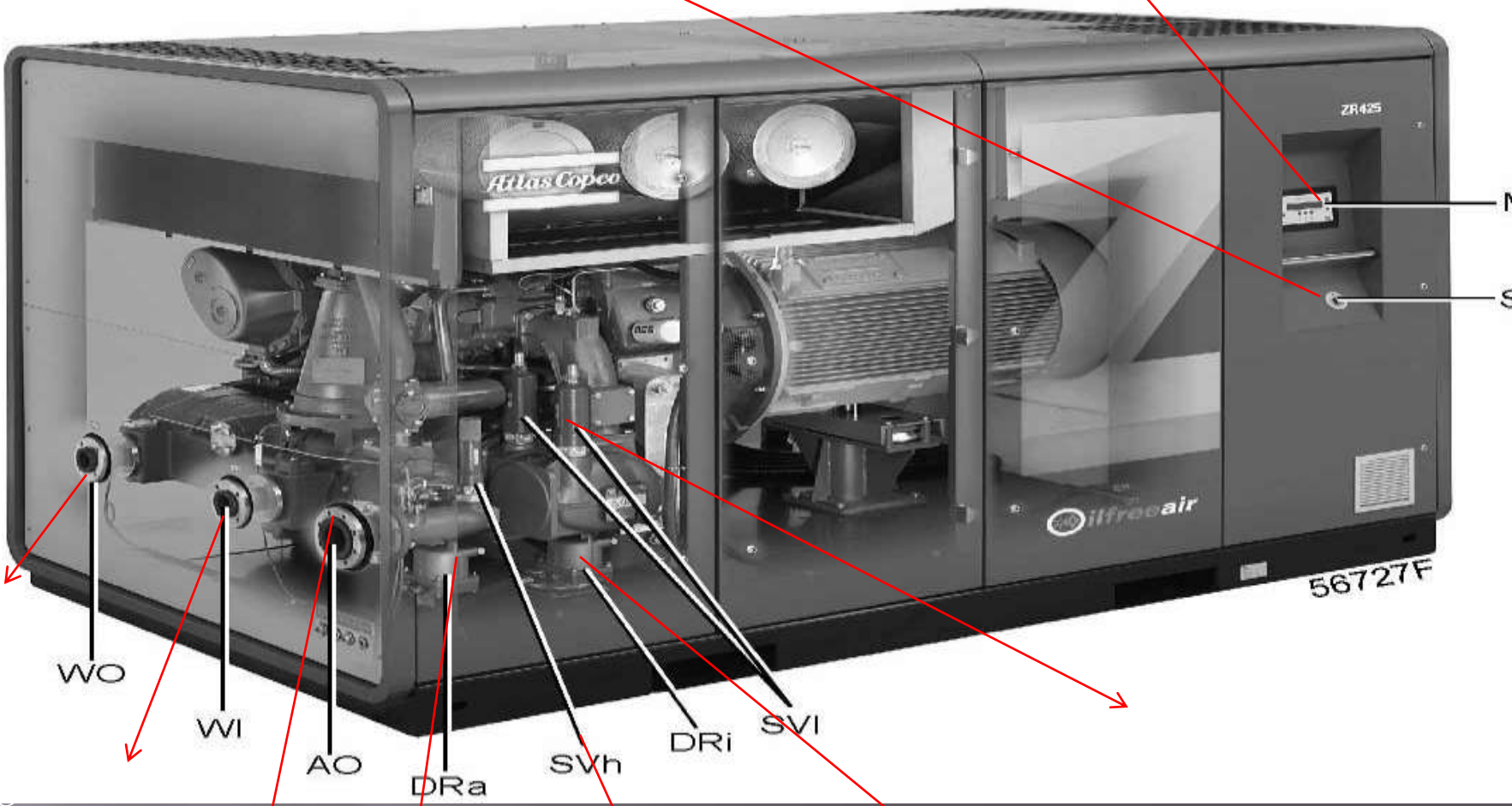
-two instrument air receivers and one ▣ separate instrument air dryers with a dew point analyzer for -40 C and two after filters need of ammonia plant

Electric-motor driven stationary compressors for industrial and mining application

Rotary helical screw compressors of this types are characterized by compact, low vibration, simple foundation, broad pressure and capacity ranges, and long life with minimum maintenance

emergency stop button

elektronikon regulator



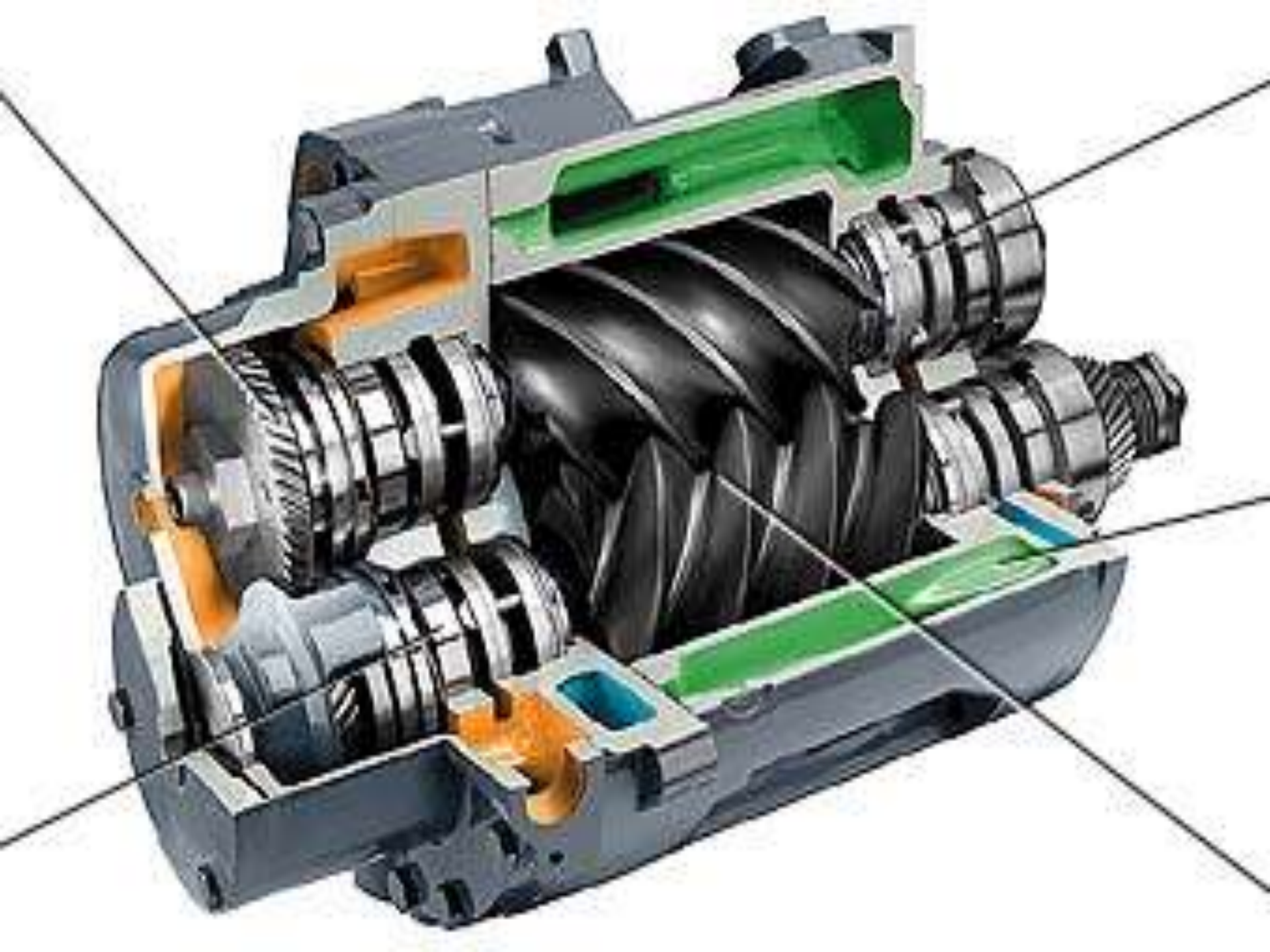
compressed air outlet

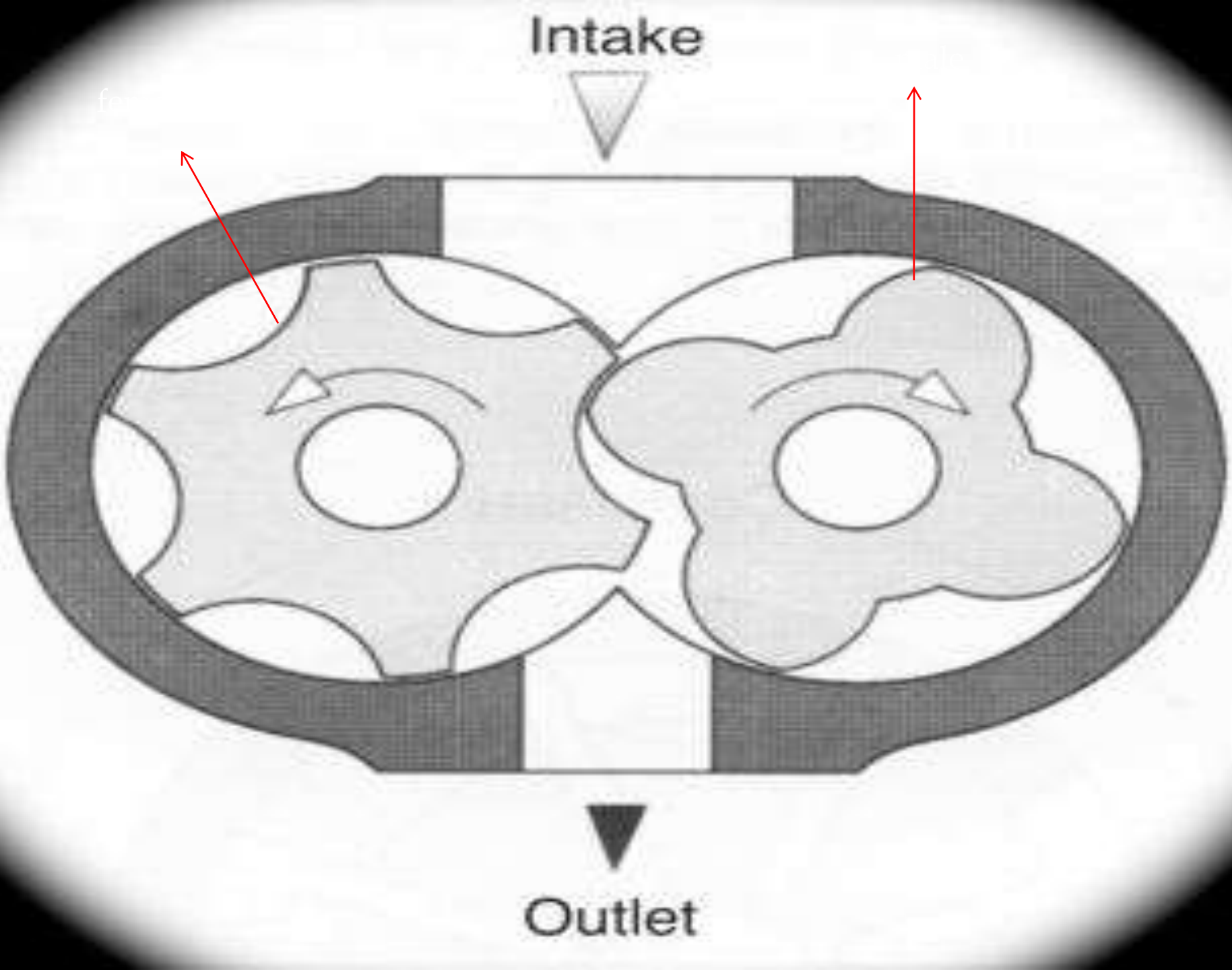
after cooler

safety valve, high pressures

inter cooler

Within the compressor body there are two screws with a female and a male screw, female having concave inlets and the male with convex helical inlets. The screws rotate in opposite directions with the female screw receiving the driving power and transmitting this power to the male screw through a set of synchronization gears.




Intake
Outlet

These compressors are tow-stage, electric motor driven screw compressors which deliver oil free, pulsation-free air

ZR compressor is enclosed in a sound-insulated bodywork and includes mainly :

- Air filter
- low pressure element
- inter cooler
- High pressure compressor element
- After cooler
- water separator
- Drive coupling
- Gear casing
- Elektronikon control system
- Safety valves.

control system

The compressor is controlled by an electronic regulator which

- Controls the operation of the compressor.
- Reduces the power consumption.
- Warns the operator in case of an abnormal condition.

Protects the compressor.

- Monitors components subject to service.
- If activated , the regulator can automatically restart the compressor after voltage failure.

Motor temperature protection

Temperature sensors are embedded in the drive motor to warn operator if temperature becomes high and to shut down the compressor to protect the motor from overheat.

Motor bearing temperature protection

temperature sensors are embedded in the drive motor to worn the operator if the temperature of the bearing becomes high and to shut down the compressor , to protect bearings from overheating.

Automatic water shut off

water shut off valve is provided in the cooling water system to shut off the system when the compressor not in operation.

Duplex oil filters

Duplex oil filters have a flow selector valve. When the handle of the valve is in the upright position, both cartridges are connected in oil circuit. When the handle is turned to the right, the left cartridge is out of service and vice versa.

Electronic drain

The EWD (Electronic Water Drain) assures proper draining of condensate and prevents water from entering the compressed air net.

SPM

A number of vibration sensors are provided in the drive motor and the compressor elements. reading can be called up on the display of the Elektronikon regulator.

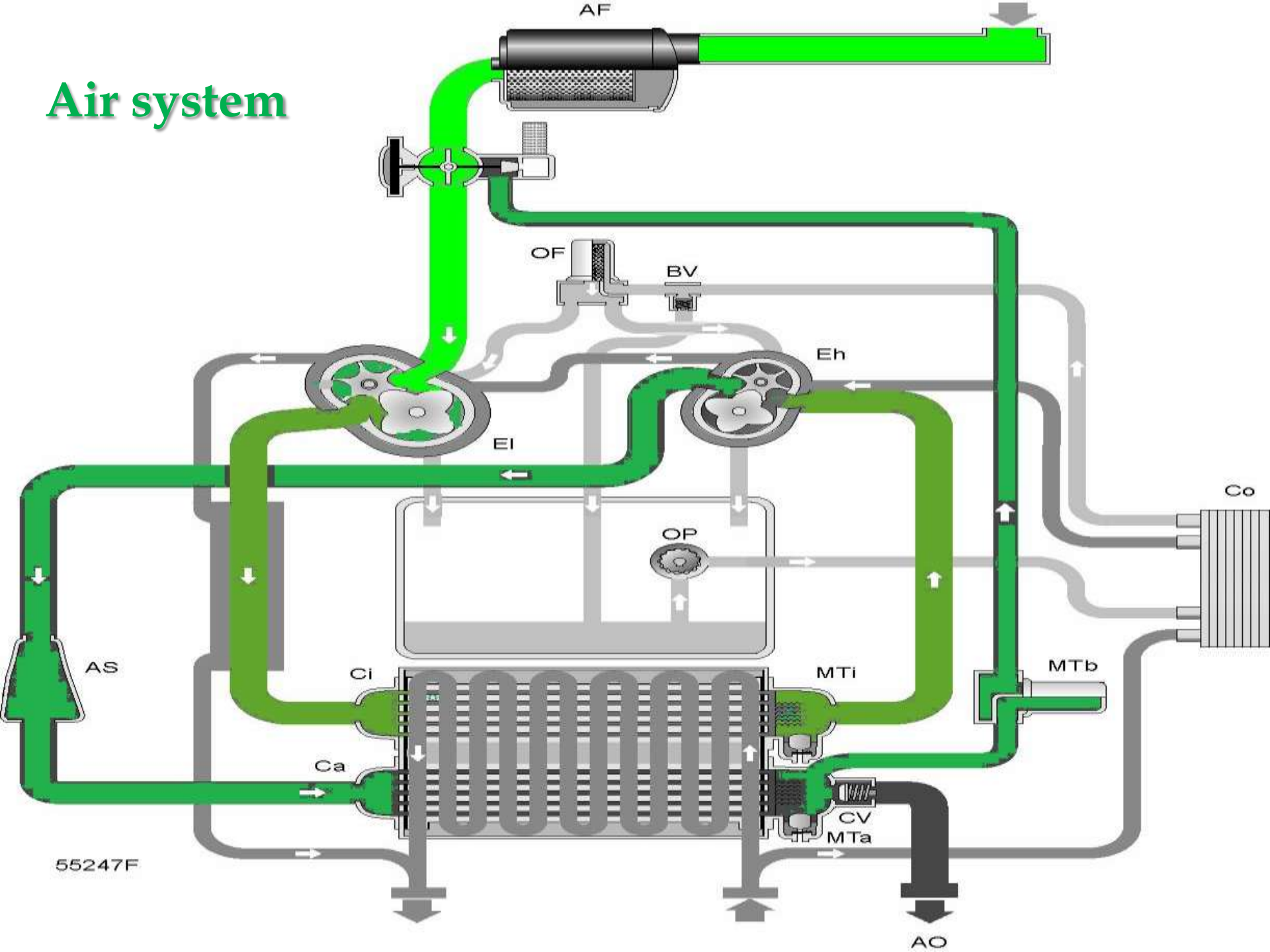
AIR SYSTEM

Air drawn through inlet filter is compressed in low pressure element (El) . The compressed air is discharged to the intercooler (Ci) .

The cooled air is farther compressed in high pressure element (Eh) . And discharged through silencer
(AS) and after cooler (Ca) a check valve (CV) is fitted downstream of the after cooler.

The compressed air leaves the compressor via the air outlet.

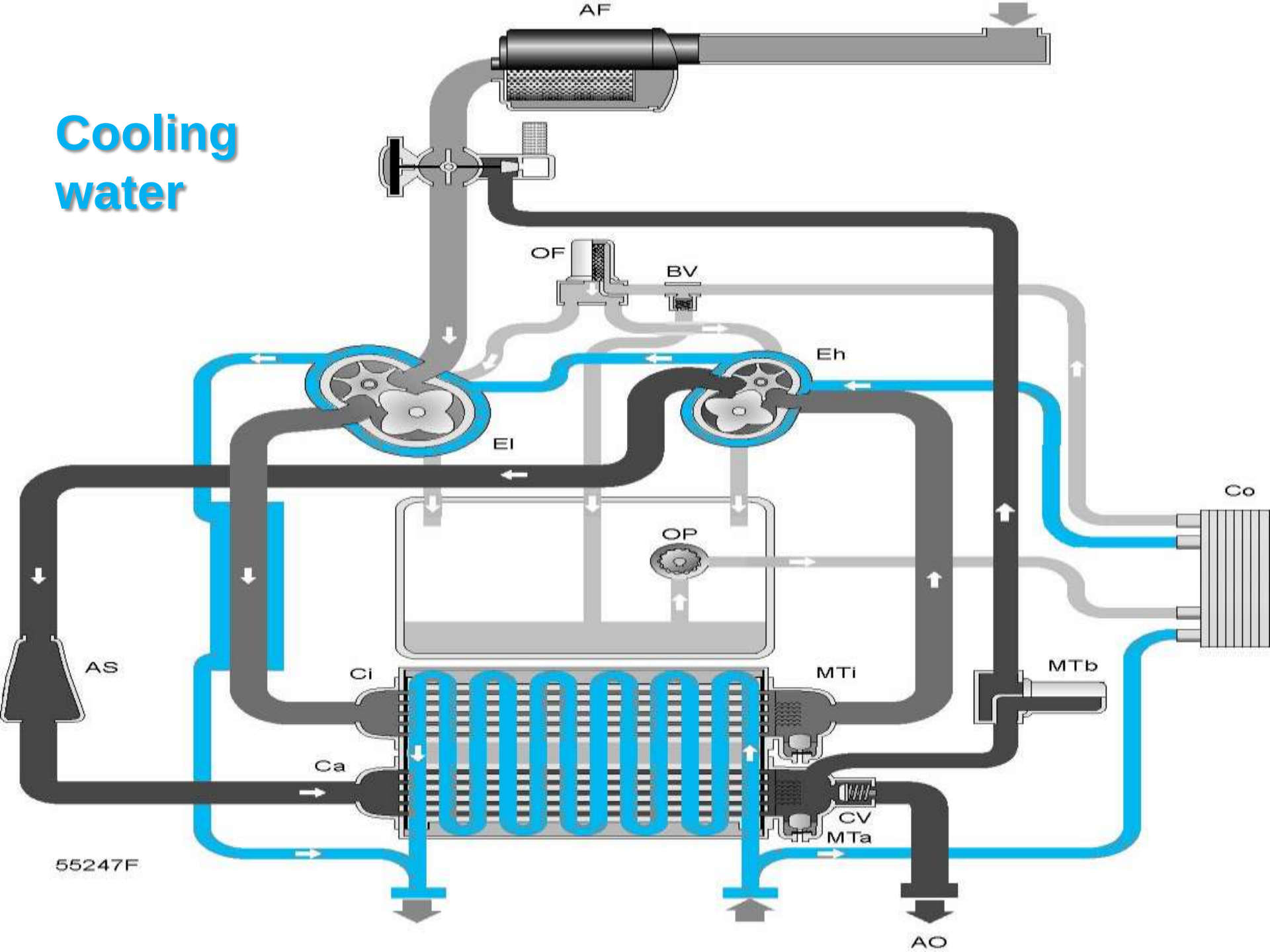
Air system



Cooling and condensate system

The cooling water flows through oil cooler , the cooling jackets of high-pressure compressor element and low-pressure element , as well as through intercooler and after cooler. The cooling water flows also around the outlet pipe of compressor element.

Cooling water



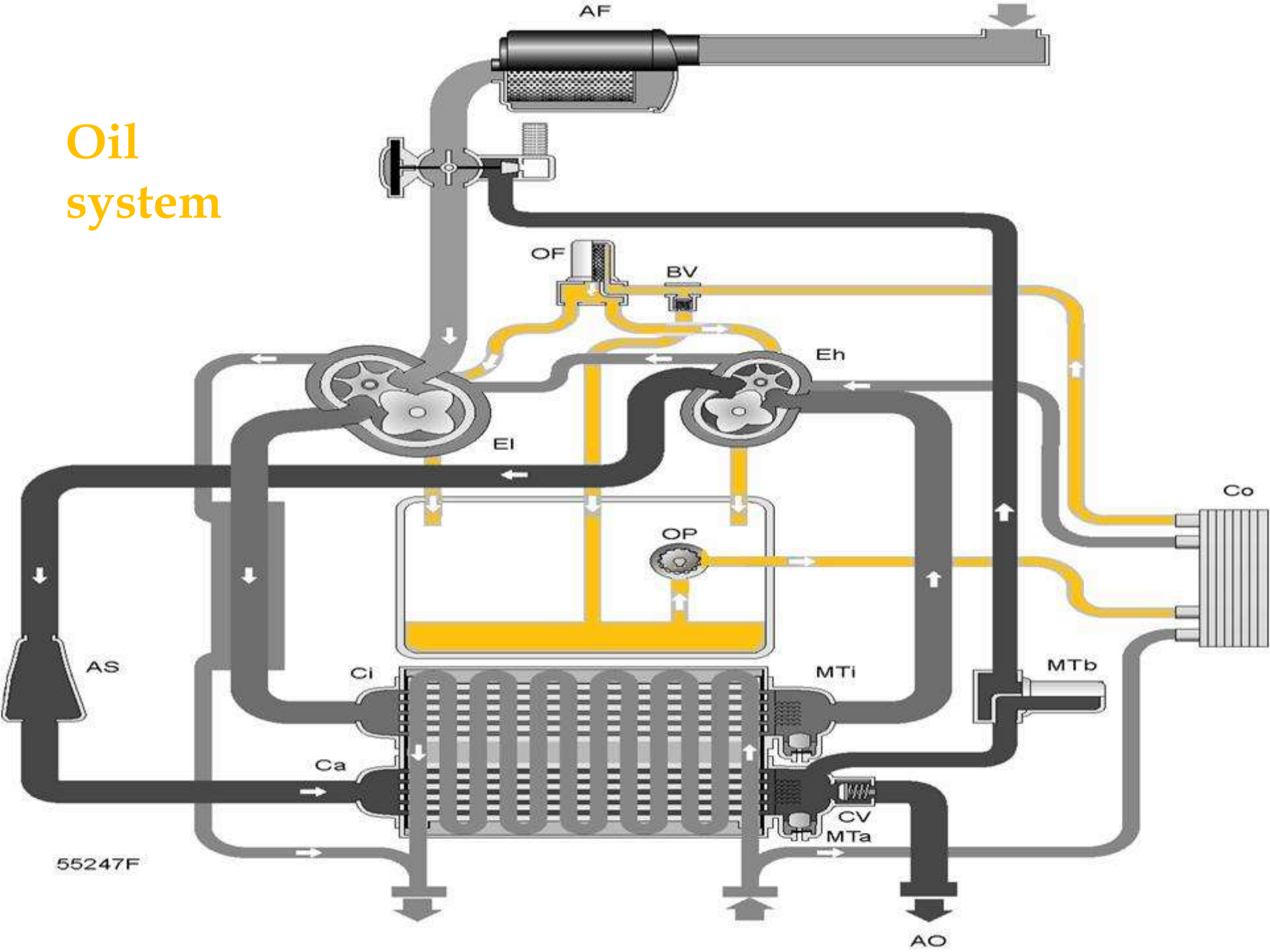
Condensate drain

Tow moisture traps are installed :
one downstream of intercooler to prevent condensate from entering the air outlet pipe. The moisture traps are connected to condensate drain receivers . Each drain receiver is fitted with a float valve to automatically drain condensate and a manual drain valve. Electronic Water Drains are available as an option. An extra moisture trap is fitted in the blow-off pipe to prevent moisture from blowing through silencers draining the transition from loaded to unloaded operation.

Oil system

Oil is circulated by pump (OP) from the sump of the gear casing through cooler (CO).
The oil passes through oil filters (OF) towards the bearings and timing gears.
Valve (BV) opens if the oil pressure should rise above a given value.

Oil system



Regulating system

The regulator keeps the net pressure within programmable pressure limits by automatically loading and unloading the compressor depending on the air consumption.

During unloading

if the air consumption is less than the air delivery of the compressor, the net pressure increases .when the net pressure reaches the upper limit of the working pressure (unloading pressure). Air delivery is stopped (0 %) , the compressor runs unloaded.

During loading

when the net pressure decreases to the lower limit of the working pressure (loading pressure). Air delivery is resumed (100) , the compressor runs loaded.

If the unloading pressure is programmed at 7.0 bar , the maximum limit for the loading pressure is changes at 6.9 bar. The recommended minimum pressure difference between loading and unloading is 0.6 bar.

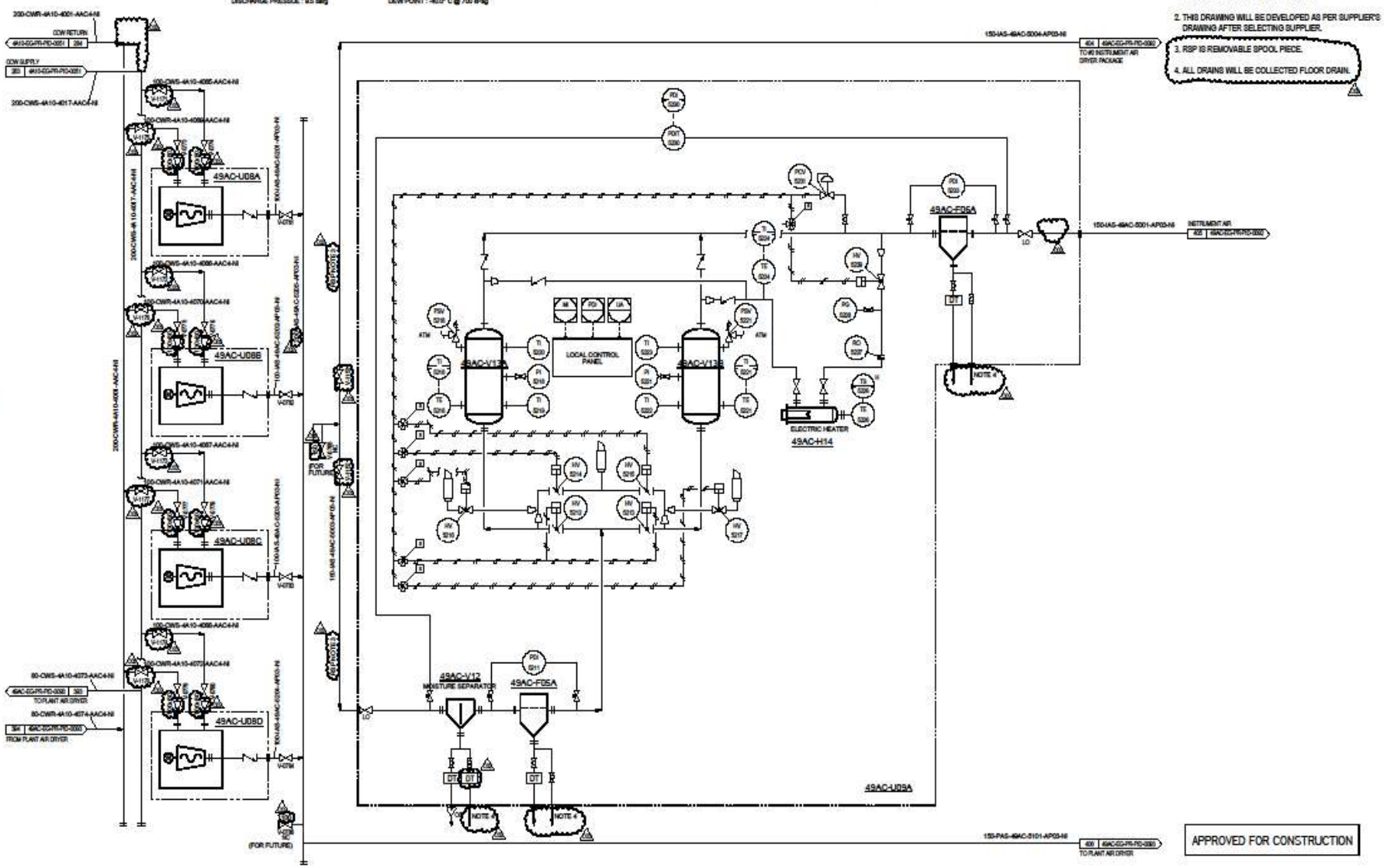
		Minimum setting	Factory setting	Maximum setting
Compressor outlet pressure (warning)	bar	0	14	17
Compressor outlet pressure (shut down)	Bar	0	15	17
Oil pressure (shut-down warning level)	Bar	1.0	1.3	1.9
Delay at start , oil pressure	Sec	15	15	20
Compressor designed for a maximum inlet temperature of 40 ˚c	˚C	100	210	220
Compressor element 1 outlet temperature (warning level)	˚C	100	225	235
Compressor element 1 outlet temperature (shut down level)	˚C	100	235	235
Compressor element 2 inlet temperature (shut-down warning level)	˚C	40	65	80

C02 | MD-502-49AC-EG-PR-PID-0091

49AC-U08A/B/C/D	49AC-U09A	49AC-V12	49AC-V13A/B	49AC-F05A	49AC-F06A	49AC-H14
AIR COMPRESSOR	#1 INSTRUMENT AIR DRYER PACKAGE	MOISTURE SEPARATOR	INSTRUMENT AIR DRYER	PRE FILTER	AFTER FILTER	ELECTRIC HEATER
TYPE : SCREW QUANTITY : 4 x 33% CAPACITY : 2007 Nm³/h DISCHARGE PRESSURE : 9.5 barg	TYPE : HEATED EXTERNAL QUANTITY : 2 x 100% CAPACITY : 4700 Nm³/hr (WET) DEW POINT : -20.0°C @ 10.1 barg	CAPACITY : 4700 Nm³/hr		TYPE : CARTRIDGE MICRO RATING : 0.01 µm	TYPE : CARTRIDGE MICRO RATING : 0.3 µm	CAPACITY : LATER

NOTES

1. FOR SYMBOL & LEGEND REFER TO P&ID 4000-EG-PR-PID-0001 THRU. 0004.
2. THIS DRAWING WILL BE DEVELOPED AS PER SUPPLIER'S DRAWING AFTER SELECTING SUPPLIER.
3. RSP IS REMOVABLE SPOOL PIECE.
4. ALL DRAINS WILL BE COLLECTED FLOOR DRAIN.



APPROVED FOR CONSTRUCTION

REFERENCE DOCUMENTS		REV	DATE	REVISION DESCRIPTION	DSN	CHK	ENG	APP	MD
4000-EG-PR-PID-0001~0004	PROCESS FLOW DIAGRAM	C00	29/12/20	REVISE TO INCORPORATE DCN-0001~0002	SHR	JKS	YJK	THC	
4000-EG-PR-MB-0101~0105	HEAT BALANCE DIAGRAM	C01	29/12/20	ISSUE FOR CONSTRUCTION	SHR	JKS	YJK	THC	

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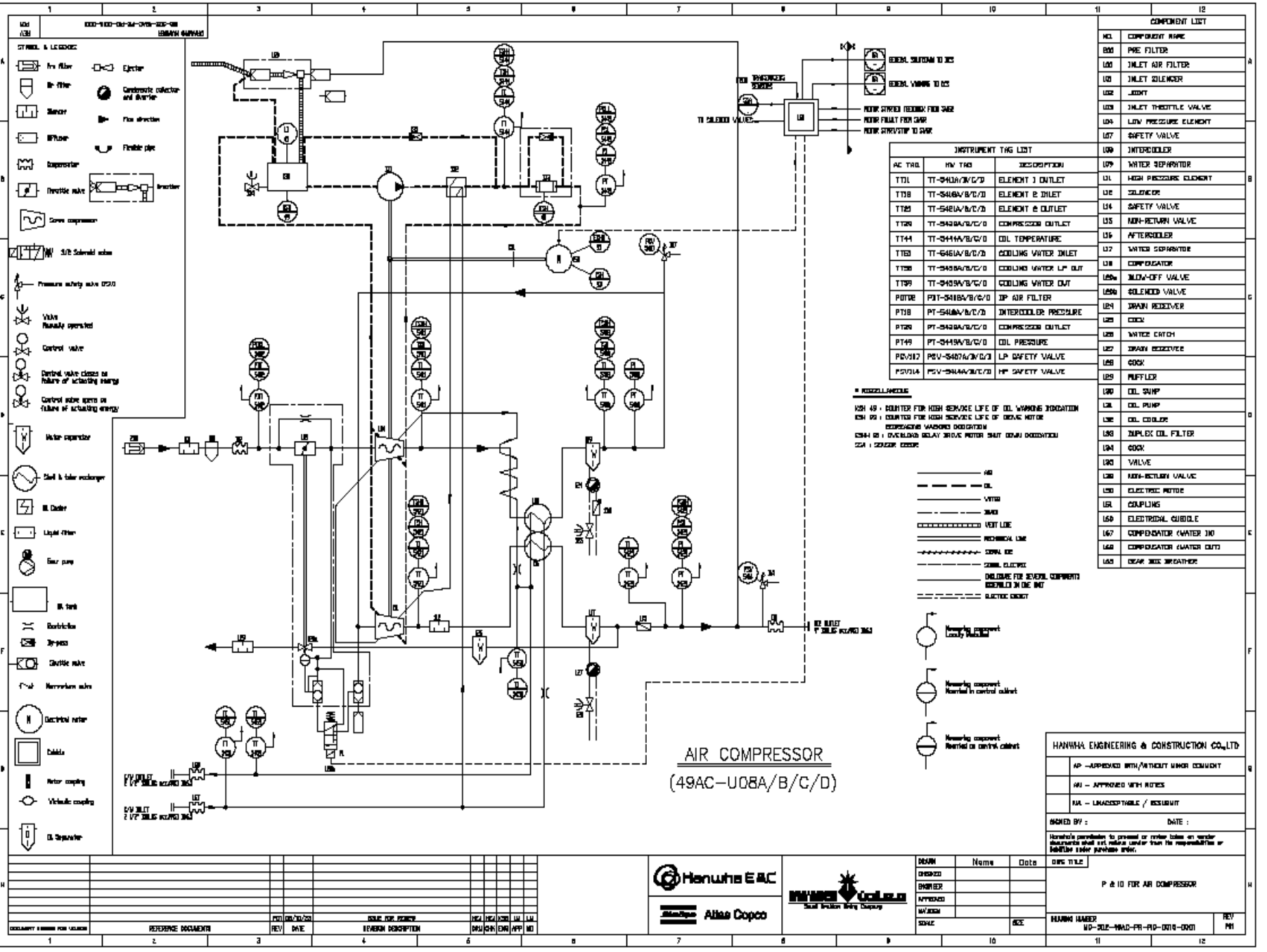
Hanwha E&C

Maaden Phosphate Company

DRAWN	DHR	29/12/20
CHECKED	JKS	29/12/20
ENGINEER	YJK	29/12/20
APPROVED	THC	29/12/20
MANAGER		
SCALE	NTS	A1

MA'ADEN PHOSPHATE PROJECT
POWER AND DESALINATION PLANT
COMPRESSED AIR SYSTEM(1/3)
PIPING & INSTRUMENTATION DIAGRAM

DRAWING NUMBER: MD-502-49AC-EG-PR-PID-0091 | REV: C02



COMPONENT LIST	
NO.	COMPONENT NAME
B00	PRE FILTER
L00	INLET AIR FILTER
L01	INLET SILENCER
L02	JOINT
L03	INLET THROTTLE VALVE
L04	LOW PRESSURE ELEMENT
L07	SAFETY VALVE
L09	INTERCOOLER
L09	WATER SEPARATOR
L11	HIGH PRESSURE ELEMENT
L12	SILENCER
L14	SAFETY VALVE
L15	NON-RETURN VALVE
L16	AFTERCOOLER
L17	WATER SEPARATOR
L18	COMPENSATOR
L18a	BLOW-OFF VALVE
L18b	SOLENOID VALVE
L24	DRAIN RECEIVER
L25	COCK
L28	WATER CATCH
L27	DRAIN RECEIVER
L68	COCK
L29	MUFFLER
L30	OIL SUMP
L31	OIL PUMP
L32	OIL COOLER
L33	DUPLEX OIL FILTER
L34	COCK
L35	VALVE
L36	NON-RETURN VALVE
L50	ELECTRIC MOTOR
L51	COUPLING
L60	ELECTRICAL CUBICLE
L67	COMPENSATOR (WATER IN)
L68	COMPENSATOR (WATER OUT)
L65	GEAR BOX BREATHER

INSTRUMENT TAG LIST		
AC TAG	HV TAG	DESCRIPTION
TT01	TT-5411A/B/C/D	ELEMENT 1 OUTLET
TT08	TT-5418A/B/C/D	ELEMENT 2 INLET
TT25	TT-5421A/B/C/D	ELEMENT 2 OUTLET
TT29	TT-5429A/B/C/D	COMPRESSOR OUTLET
TT44	TT-5414A/B/C/D	OIL TEMPERATURE
TT63	TT-5461A/B/C/D	COOLING WATER INLET
TT58	TT-5458A/B/C/D	COOLING WATER LP OUT
TT59	TT-5459A/B/C/D	COOLING WATER OUT
PDT08	PDT-5418A/B/C/D	IP AIR FILTER
PT08	PT-5418A/B/C/D	INTERCOOLER PRESSURE
PT29	PT-5429A/B/C/D	COMPRESSOR OUTLET
PT49	PT-5449A/B/C/D	OIL PRESSURE
PSV017	PSV-5407A/B/C/D	LP SAFETY VALVE
PSV014	PSV-5414A/B/C/D	HP SAFETY VALVE

* FUNCTIONING

KSH 49 : COUNTER FOR HIGH SERVICE LIFE OF OIL WARNING INDICATION
KSH 99 : COUNTER FOR HIGH SERVICE LIFE OF DRIVE MOTOR BEARINGS WARNING INDICATION
K5H4-95 : OVERLOAD RELAY DRIVE MOTOR SHUT DOWN INDICATION
SSA : SENSOR ERROR

HANWHA ENGINEERING & CONSTRUCTION CO.,LTD	
AP – APPROVED WITH/WITHOUT MINOR COMMENT	
AN – APPROVED WITH NOTES	
NA – UNACCEPTABLE / RESUBMIT	
SIGNED BY :	DATE :

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	Name	Date
DRAWN		
DESIGNED		
ENGINEER		
APPROVED		
MANAGER		
SCALE		SIZE

DWG TITLE: P & ID FOR AIR COMPRESSOR

HANWHA NUMBER: HD-SOLE-HMAC-PR-PID-0016-0001 REV: M1

REV	DATE	ISSUE FOR REVIEW / REVISION DESCRIPTION	DRAWN	CHECKED	ENGINEER	APPROVED	MANAGER
M1	08/10/20	ISSUE FOR REVIEW	HCJ	GHK	KSB	LJH	LJH

DOCUMENT NUMBER FOR VENDOR | REFERENCE DOCUMENT

Hanwha E&C | Atlas Copco | HANWHA Qcells Social Solution Driving Company

ADSORPTION AIR DRYER

Description

The de-humidification in compressed air is the most important matter in all the fields
using compressed air. Because the moisture of compressed air badly affects Industrial manufacturing processes and instrumentation equipment.
To obtain the compressed dry air, you can use Heater External Type dryer which is excellent in not being polluted and safety. This Heater External Type dryer has excellent performance in comparison with a heatless type dryer which has been widely used.

The general functional features of this Heater External Type dryer are as follows.

1) Long life of Desiccant – Because the heater is designed as external-type and thus regenerated by the indirect heat contact of the desiccant, the life of the desiccant is long compared with a heatless type dryer.

2) Prevention of a dew point fall – Completely prevents the phenomenon that the dew point sharply falls at the time of Tower switching by regeneration of heat source of the heater compared with a heatless type dryer.

3) Super low dew point – Because dry air is heated to use for regeneration, this drying system is most suitable for the needs of super low dew point and at present is widely used.

4) ELECTRIC HEATER – The heater is controlled by setting the temperature suitable for regenerating the desiccant, so unnecessary electric power does not waste. Also, to ensure safe operation, this dryer has a safety device which is automatically turned off when ELECTRIC HEATER is overheated.

5) System application – Heater External Type dryer can apply an energy saving type dryer system and has excellent performance. Application example: Combination Type dryer, Pulse Purge Regeneration System.

C02 | MD-502-49AC-EG-PR-PID-0092

49AC-U09B
#2 INSTRUMENT AIR DRYER PACKAGE
TYPE : HEATLESS/EXTERNAL
QUANTITY : 1 SET
CAPACITY : 3x1200 Nm³/h(MAX.5121 Nm³/h)
DEW POINT : -40°F @ 100 kPag

49AC-V22
MOISTURE SEPARATOR
CAPACITY : 4200 Nm³/hr

49AC-V23A/B
INSTRUMENT AIR DRYER

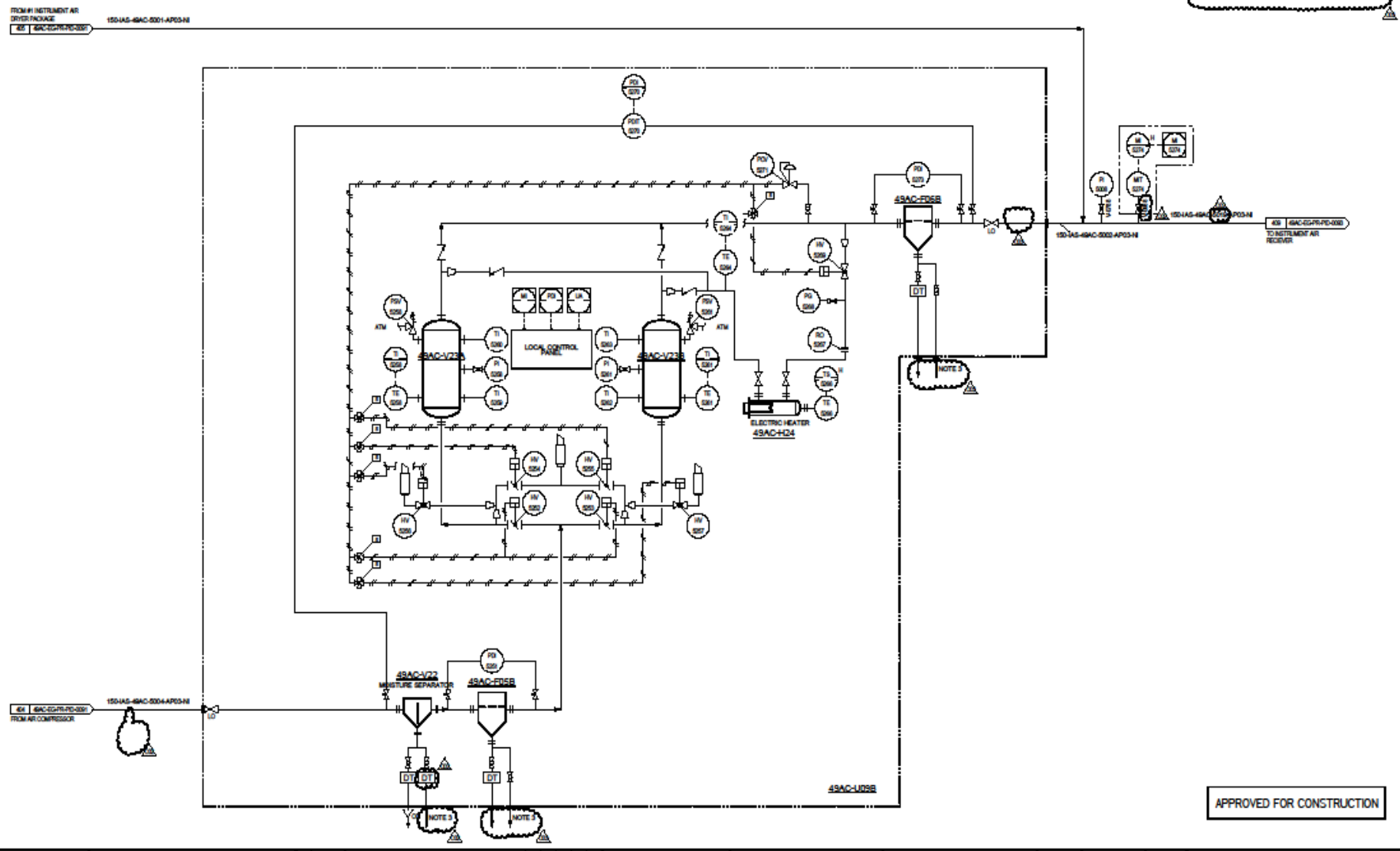
49AC-F05B
PRE FILTER
TYPE : CARTRIDGE
QUANTITY : 1 x 100%
MICRO RATING : 0.01 μm

49AC-F06B
AFTER FILTER
TYPE : CARTRIDGE
QUANTITY : 1 x 100%
MICRO RATING : 0.3 μm

49AC-H24
ELECTRIC HEATER
CAPACITY : LATER

NOTES

1. FOR SYMBOL & LEGEND REFER TO P&ID 4000-EG-PR-PID-0001 THRU. 0004.
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3. ALL DRAINS WILL BE COLLECTED FLOOR DRAIN.



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REFERENCE DOCUMENTS		REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	MD
4000-EG-PR-PID-0001~0004	PROCESS FLOW DIAGRAM	C02	26/12/08	REVISE TO INCORPORATE DCN-0001~0002	SHK	JKS	YJK	THC	
4000-EG-PR-MEB-0101~0105	HEAT BALANCE DIAGRAM	C01	26/09/08	ISSUE FOR CONSTRUCTION	SHK	JKS	SWC	HAM	

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Hanwha E&C

Maaden Phosphate Company

DRAWN	SHK	26/12/08
CHECKED	JKS	26/12/08
ENGINEER	YJK	26/12/08
APPROVED	THC	26/12/08
MANAGER		
SCALE	NTS	A1

MA'ADEN PHOSPHATE PROJECT
POWER AND DESALINATION PLANT
COMPRESSED AIR SYSTEM(2/3)
PIPING & INSTRUMENTATION DIAGRAM

DRAWING NUMBER: MD-502-49AC-EG-PR-PID-0092 | REV: C02

EXPLANATION OF THE SYSTEM

DRYING

The air compressed by the compressor enters the first tower on one side through 3-way shuttle v/v or 2-way butterfly v/v located in the inlet of the dryer.

This air passes through the desiccant within the tower and goes toward the upper part of the tower, when the moisture of particles in the air is removed and the air becomes dry, discharged toward the outlet. This phenomenon is called as Drying Cycle.

- Standard KHP type drying cycle: 4.0hr

Regeneration

Some of the dry air discharged toward the outlet is put into the tower on the opposite side
to dehumidify and clean the desiccant and is exhausted into the air through the muffler.
This is called as regeneration, which is composed of two processes

1) Heating

The air is heated by an electric heater before entering the regeneration tower, which is called as heating. ELECTRIC HEATER is automatically operated and turned and off by T.I.C. (temp. indicating controller) on the control panel.

- Standard KHP Type heating cycle: 2.5 hr.

1) Heating

The air is heated by an electric heater before entering the regeneration tower, which is called as heating. ELECTRIC HEATER is automatically operated and turned and off by T.I.C. (temp. indicating controller) on the control panel.

- Standard KHP Type heating cycle: 2.5 hr.

2) Cooling

When timer heating time (2.5 hr) on the control panel is reached, the power of ELECTRIC HEATER is automatically cut off and the air toward the outlet not heated comes in to cool the regeneration tower, which is called as cooling.

- Standard KHP Type cooling cycle: 1 hr 25 min ~ 1 hr 27 min.

PRESSURIZING

After cooling cycle is finished, the purge valve toward the regeneration tower is closed
and then the pressure of the regeneration tower is gradually higher and at last the
pressure became the same as that of the drying tower, which is called as pressurizing.

- Standard KHP Type pressurizing cycle: 3 min ~ 5 min

After enough pressurizing is finished, the depressurizing purge v/v toward the drying tower is opened and the air of the drying tower is slowly exhausted into the air, which is called as depressurizing purge (First purge) and the purge time is set by a separate timer.

Note) This is installed by the customer’s option or special specifications, and in case of standard products, it is not usually installed except for some models.

MAIN PURGE

After enough pressurizing is finished, the purge valve toward the drying tower is opened and the pressure of the tower is discharged into the air, which is called as purge.

In case DEPRESSURIZING PURGE V/V was installed, main purge is in progress after the depressurizing purge.

FUNCTION OF COMPONENTS

DRYING TOWER

There are two towers in “KHP” Heater External Blower Type Dryer and a desiccant is mounted in the inside of the tower, so the dryer supplies dry air by absorbing the moisture of the compressed air passing through the inside of the drying tower.

A stainless screen was installed in the nozzles of the inlet and outlet of the upper and lower parts so that the grains of the desiccant may not flow in the piping line, and each nozzle is attached for mounting and discharge of the desiccant.

Also, only the tower which passed the design and performance inspection of Korea Occupational Safety & Health Agency is used.

AIR INLET V/V

This let the compressed air that flows in “KHP” Dryer go toward A or B tower, and the operation cycle is automatically switched in accordance with the time set in the timer, and in case of a standard product, it is manufactured by the following table. In case 2-way butterfly v/v is used, it operates the actuator by using a separate solenoid v/v.

PURGE V/V

This operates in the process of regeneration and pressurization, exhausts regeneration air into the air, and the operation cycle is automatically adjusted in accordance with the set time. It starts the actuator by using a separate actuator.

DEPRESSURIZING PURGE V/V

This operates in the process of regeneration and pressurization, firstly exhausts the regeneration air into the air before operation of PURGE V/V and its valve is opened/closed by a separate depressurizing timer. (Applies to only some models)

AIR OUTLET CHECK VALVE

The compressed air which passed through the desiccant in the drying tower and whose moisture was removed enters the line after passing through the valve of the upper pipe of the tower. This valve, called as AIR OUTLET CHECK VALVE, let the air move only in one direction, thus preventing the drying air from being mixed with the regeneration air.

PURGE AIR CHECK VALVE

This valve let the air flow in the tower on the opposite side of the drying tower by using some compressed air whose moisture was removed. At this time, the air passes through PURGE AIR CHECK VALVE into the tower, and this check valve let the air move only in one direction, thus preventing the regeneration air being mixed with the purge air.

PURGE ADJUSTING VALVE

Before the cooling air flows in the generation tower, the flow rate of the cooling air is adjusted suitably for generation through this valve by the pressure of PURGE FLOW PRESS. GAUGE. (About 2.5 ~ 3.5 kg/㎠).

PURGE SHUT OFF VALVE

This valve is closed just before depressurizing purge begins after pressurizing, and let the regeneration air exhaust into the air when depressurizing purge is in progress. This is opened when purge is finished. (apply to only some models).

ELECTRIC HEATER

This is used for generating a desiccant by heating regenerated air and 1 set is installed in the inside of the dryer base.

In case of regenerated air, the dew point of the outlet is affected by the regeneration degree of the desiccant by heating temperature, so if the heating temperature is lower than the set temperature, the performance can’t be ensured.

The temperature of ELECTRIC HEATER is adjusted on T.I.C of the panel front door and a separate thermo switch for preventing

overheating of ELECTRIC HEATER was installed in the inside of the panel.
T.I.C (temp. indicating controller) set value: 180℃ ~ 250℃.
330℃ Note) The above set value may be changed according to an operation situation.

PRESSURE GAUGE AND TEMPERATURE GAUGE

The pressure gauge and temperature gauge was installed in the pipe of the dryer and drying tower for ascertaining an operation situation. It is necessary to operate the machine paying attention to the heating temperature especially in the heating process of the regenerated tower. If the heating temperature becomes low, it has a bad effect on the regeneration of a desiccant, resulting in performance being worse.

DEW POINT METER

Installed in the panel of the dryer, this indicates the dew point of the outlet .

THANK YOU